

Test Cases — DNS Zone Reviewer Bot

Summary

Total Tests	Passed	Failed	Coverage
50	50	0	Core pipeline

Test Suite 1 — DNS Validator (test_validator.py)

Test Name	What it tests	Expected Result
TestValidCleanZone		
test_valid_returns_true	Validating a clean zone file	valid=True returned
test_no_errors	Clean zone file parse errors	errors=[] returned
test_no_findings	Clean zone file findings	findings=[] returned
test_minimal_valid_zone	Minimal valid zone with SOA, NS, A	valid=True and zero findings
TestWildcardRecord		
test_wildcard_inline_is_critical	Zone with a wildcard A record	At least one CRITICAL finding
test_wildcard_finding_record_type	CRITICAL finding references correct type	Finding references 'A' record and 'wildcard'
test_test_domain_has_wildcard_critical	Canonical test domain with wildcard	CRITICAL finding for wildcard
TestMissingNS		
test_missing_ns_is_critical	Zone missing an NS record	CRITICAL finding for 'NS'
test_test_domain_missing_ns_critical	Canonical test domain missing NS	CRITICAL finding for 'NS'
TestMissingSOA		
test_missing_soa_is_critical	Zone missing an SOA record	CRITICAL finding for 'SOA'
TestLowTTLMX		
test_low_ttl_mx_inline_is_warning	MX record with TTL < 300	WARNING finding for 'MX'

Test Name	What it tests	Expected Result
test_low_ttl_mx_description_mentions_ttl	WARNING description text	Description includes TTL value '60'
test_test_domain_low_ttl_mx_warning	Canonical test domain low-TTL MX	WARNING finding for 'MX'
test_normal_ttl_mx_no_warning	MX record with normal TTL	No TTL warning for 'MX'
TestLowTTLA		
test_low_ttl_a_is_warning	A record with TTL < 300	WARNING finding for 'A'
TestInvalidZone		
test_invalid_zone_returns_valid_false	Garbled/corrupted zone file	valid=False returned
test_invalid_zone_has_errors	Invalid zone file error messages	Errors list is not empty
test_invalid_zone_has_no_findings	Invalid zone file findings list	findings=[] returned
test_empty_zone_missing_required_records	Empty zone file missing SOA/NS	CRITICAL findings for 'SOA' and 'NS'
TestDifferentialChecks		
test_soa_serial_regression_is_critical	Decreasing SOA serial number	CRITICAL finding for 'SOA'
test_new_txt_record_produces_info	Adding a new TXT record	INFO finding for 'TXT'
test_ttl_change_produces_info	Changing an A record's TTL	INFO finding mentioning 'ttl'

Test Suite 2 — LLM Reviewer (test_llm_reviewer.py)

Test Name	What it tests	Expected Result
test_successful_llm_response	Successful LLM response from API	Parsed response matches expected valid LLM data
test_fallback_unreachable	Graceful fallback when API is unreachable	llm_available=False, safe_to_merge=True, proper fallback findings
test_fallback_invalid_json	Fallback when LLM returns invalid JSON	llm_available=True, risk_level="MEDIUM", parsed fallback findings
test_fallback_fields_present	Fallback response format completeness	All required fields present in the output

Test Suite 3 — GitHub Poster

(test_github_poster.py)

Test Name	What it tests	Expected Result
TestBuildHeaders		
test_headers_contain_bearer_token	Authorization header	Header contains Bearer <TOKEN>
test_headers_contain_accept	Accept header	Header contains application/vnd.github+json
test_headers_contain_api_version	X-GitHub-API-Version header	Header contains 2022-11-28
TestSuccessfulPost		
test_post_returns_success	HTTP 201 behavior	success=True, post called once
test_comment_url_returned	Comment URL extraction	Returned comment_url matches API response
test_post_sends_correct_payload	POST request payload	Request body exactly matches comment text
TestDeleteExistingBotComment		
test_deletes_previous_bot_comment	Handling existing bot comments	DELETE API called on previous comment ID
test_no_delete_when_no_bot_comment	Handling no previous bot comments	DELETE API is not called
TestFindExistingBotComment		
test_finds_bot_comment	Search for existing bot comment	Bot comment ID returned
test_returns_none_when_no_bot_comment	Search when no bot marker is found	None returned
test_returns_none_on_api_error	Search during GET API failure	None returned gracefully
TestMissingToken		
test_empty_token_returns_failure	Empty token validation	success=False, error mentions empty/missing token
test_empty_token_skips_api_calls	Empty token API behavior	No API calls are made
TestApiErrors		
test_403_returns_failure	HTTP 403 (insufficient permissions)	success=False, error includes "403"
test_422_returns_failure	HTTP 422 (validation failed)	success=False, error includes "422"

Test Name	What it tests	Expected Result
<code>test_network_error_returns_failure</code>	Network exception handling	<code>success=False</code> , error mentions network failure
TestFallbackPrint		
<code>test_fallback_prints_comment</code>	Fallback on API failure	Comment body prints to stdout
<code>test_missing_token_prints_fallback</code>	Fallback on missing token	Comment body prints to stdout

Test Suite 4 — Integration Tests (`test_integration.py`)

Test Name	What it tests	Expected Result
<code>test_full_pipeline_clean_zone</code>	Pipeline execution with a clean zone	<code>safe_to_merge=True</code> and risk is LOW/CLEAN
<code>test_full_pipeline_risky_zone</code>	Pipeline execution with a risky zone	<code>safe_to_merge=False</code> and findings contain CRITICAL
<code>test_full_pipeline_critical_zone</code>	Pipeline execution with a critical zone	<code>valid=False</code> or CRITICAL finding present
<code>test_comment_formatter_clean_output</code>	Formatter output for clean zone	Comment contains "CLEAN" and "Safe to Merge"
<code>test_comment_formatter_critical_output</code>	Formatter output for critical zone	Comment contains "CRITICAL" and "☐"
<code>test_main_exits_with_code_1_on_critical</code>	Main script behavior on CRITICAL risk	Script terminates with SystemExit code 1

How to Run Tests

```
pip install -r requirements.txt
python -m pytest -v
```

Test Results Screenshot

[50 tests passing — see pytest output in CI logs]